

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I VISHAL, M (192565077) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M. Vishal

19/20

ASHP

the following Questions.

Problem Understanding.

Subnet the Network 192.168.20.0/24 into:

- * Lab A : 126
- * Lab B : 127
- * Lab C : 128

Find the network address, host Range, broadcast address, and usable hosts.

Method Selection

Use CIDR Subnetting

$$* \text{Total Addressing} = 2^n (32 - \text{Prefix})$$

$$* \text{Usable Hosts} = \text{Total Address} - 2$$

Solution Process:

Lab	Network	Host Range	Broadcast	Usable Host
Lab A (126)	192.168.20.0	192.168.20.1	192.168.20.63	62
Lab B (127)	192.168.20.64	192.168.20.65-94	192.168.20.95	30
Lab C (128)	192.168.20.96	192.168.20.97-110	192.168.20.111	14

Accuracy of Calculation:

The Subnet addresses, host Ranges, Broadcast, addresses, and Usable Hosts are calculated correctly.

Interpretation of Results:

The 124 network is divided into three subnets with 62, 30, and 14, usable hosts for Lab A, B and C respectively.

the following Questions:

Problem Understanding.

Subnet the Network 192.168.20.0/24 into:

- * Lab A : 128
- * Lab B : 129
- * Lab C : 128

Find the network address, host Range, broadcast address, and usable hosts.

Method Selection

Use CIDR Subnetting

- * Total Addressing = 2^n (32 - prefix)
- * Usable Hosts = Total Address - 2

Solution Problem

Lab	Network	Host Range	Broadcast	Usable Host
Lab A (128)	192.168.20.0	192.168.20.1	192.168.20.63	62
Lab B (129)	192.168.20.64	192.168.20.65-94	192.168.20.95	30
Lab C (128)	192.168.20.96	192.168.20.97-126	192.168.20.127	14

Accuracy of Calculation:

The Subnet addresses, host Ranges, Broadcast, Addresses, and Usable Hosts are Calculated correctly

Interpretation of Results:

The 124 network is divided into three Subnets with 62, 30, and 14, usable hosts for Lab A, B and C. respectively

2.

Problem Understanding

Find the Subnet, network address, broadcast address, host range, and usable hosts for the given IP addresses using 255.255.255.192 (126)

Method Selection:

Use 126 Subnetting. Each Subnet has 64 addresses and 62 usable hosts.

Solution Process:

System 1 (192.168.1.130):

* Network: (192.168.1.128):

* Host Range: 192.168.1.129 - 192.168.1.190

* Broadcast: 192.168.1.191

System 2 (192.168.1.190):

* Network: 192.168.1.128

* Host Range: 192.168.1.129 - 192.168.1.190

* Broadcast: 192.168.1.191

Usable Hosts: 62

Accuracy of Calculation:

Both IP addresses belong to the same (192.168.1.128)

Subnet Interpretation of Results:

The two System Can Communicate directly because they are in the same Subnet.

Problem Understanding:

Subnet 172.16.0.0/16 for HR (125), Sales (126), IT (127), and Admin (128)

Method Selection:

Allocate Subnets Sequentially Using VLSM

Solution Process:

Department	Network	Broadcast
HR	172.16.0.0/26	172.16.0.127
Sales	172.16.0.0/27	172.16.0.191
IT	172.16.0.0/28	172.16.0.223
Admin	172.16.0.0/29	172.16.0.239

Accuracy of Calculation:

All Subnet addresses and Host Counts are correct

Result:

The remaining IP addresses from 172.16.0.240 onward remain available for future use.

4. Problem of Understanding:

Subnet 172.16.10.0/24 using /27 and identify the

Subnet for 172.16.10.78.

Method Selection:

A /27 Subnet provides 30 usable hosts

Solution Process:

* Total Subnets: 8

* Usable Hosts / Subnet: 30

* IP 172.16.10.78 belongs to subnet 172.16.10.64/27

* Network: 172.16.10.95

* Broadcast: 172.16.10.95

* 172.16.10.98 is the broadcast address not a host

Accuracy of calculation:

The subnet and broadcast address are correctly identified.

Result:

The given IP belongs to the 172.16.10.64/27 Subnet.

5. Problem Understanding

Compress and expand the given IPv6 address, find the /64 Network Prefix, and Calculate Host address.

Method Selection:

Apply IPv6 Compression rules and use the /64

Prefix

Solution Problem:

* Compressed: 2001:db8::f00:42:8329

* Expanded:

2001:0db8:0000:0000:0000:f00:0042:8329

* Network Prefix: 2001:db8:0:0::/64

* Host Addresses: 2^{64}

Accuracy of calculation:

The IPv6 Compression, expansion, and Network Prefix are correct.

Interpretations of Results:

A /64 IPv6 network provides 2^{64} possible host addresses, making it suitable for large-scale Network.